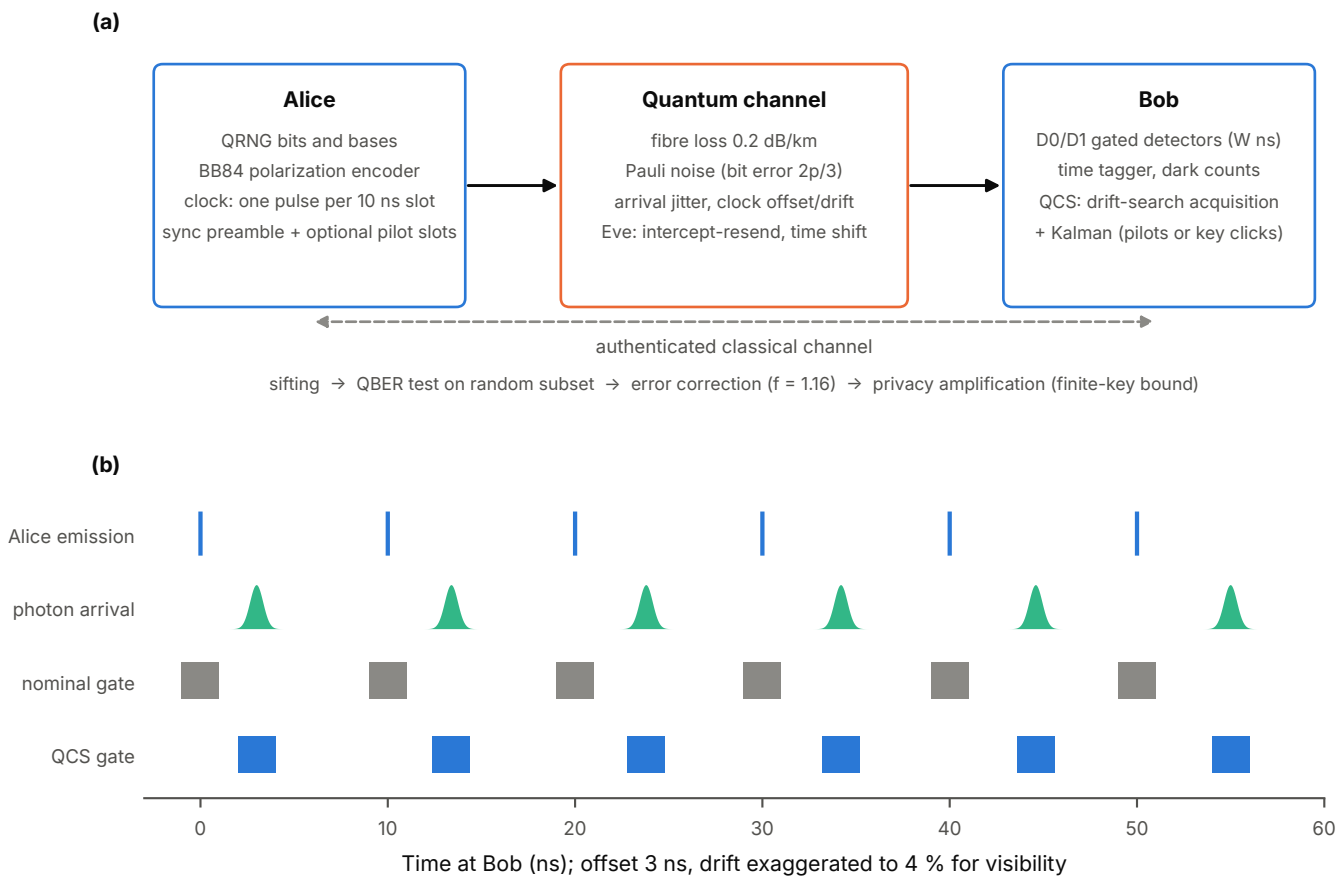
(a)
Alice
QRNG bits and bases
BB84 polarization encoder
clock: one pulse per 10 ns slot
sync preamble + optional pilot slots
Quantum channel
fibre loss 0.2 dB/km
Pauli noise (bit error 2p/3)
arrival jitter, clock offset/drift
Eve: intercept-resend, time shift
Bob
D0/D1 gated detectors (W ns)
time tagger, dark counts
QCS: drift-search acquisition
+ Kalman (pilots or key clicks)
authenticated classical channel
sifting → QBER test on random subset → error correction (f = 1.16) → privacy amplification (finite-key bound)
(b)
Alice emission
photon arrival
nominal gate
QCS gate
0
10
20
30
40
50
60
Time at Bob (ns); offset 3 ns, drift exaggerated to 4 % for visibility